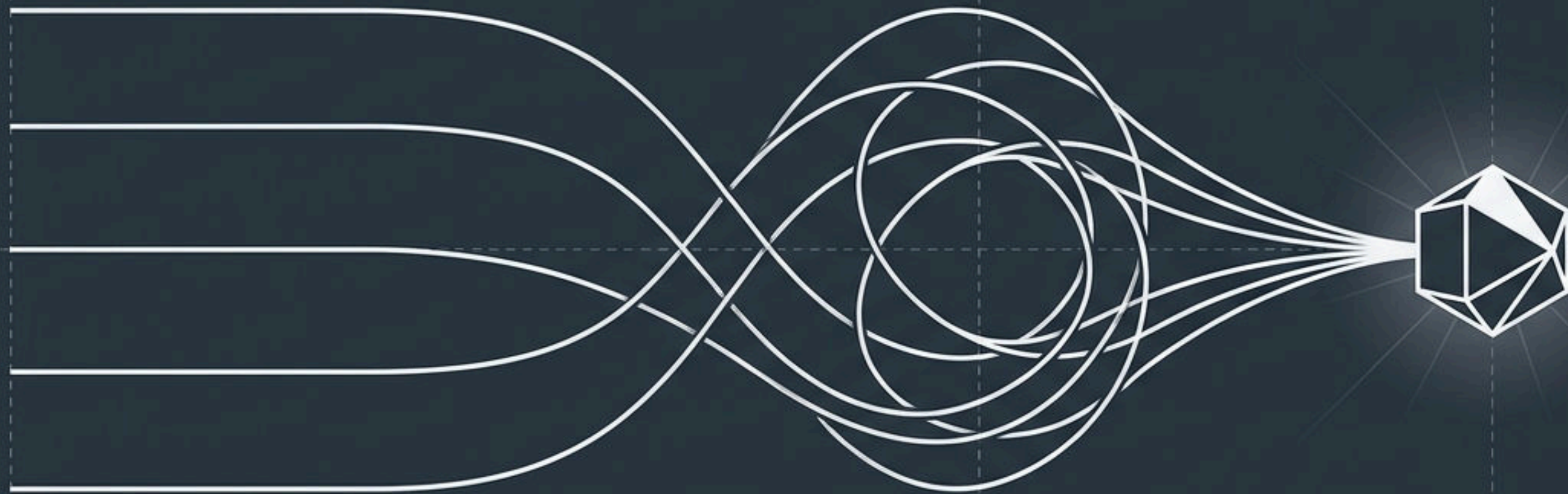


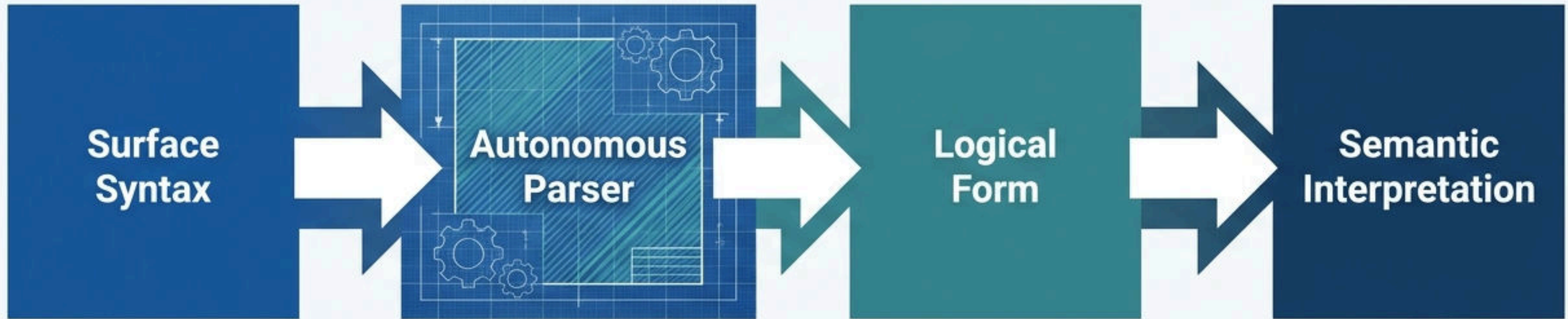


What Survives Reconstruction

Admissibility, Irreversibility, and the Architecture of Externalized Structure

Flyxion | July 2026

The Category Mistake of the Claude Code Leak



The Incident

In March 2026, the complete source of Anthropic's Claude Code CLI leaked, revealing hand-written deterministic parsers alongside the neural model.

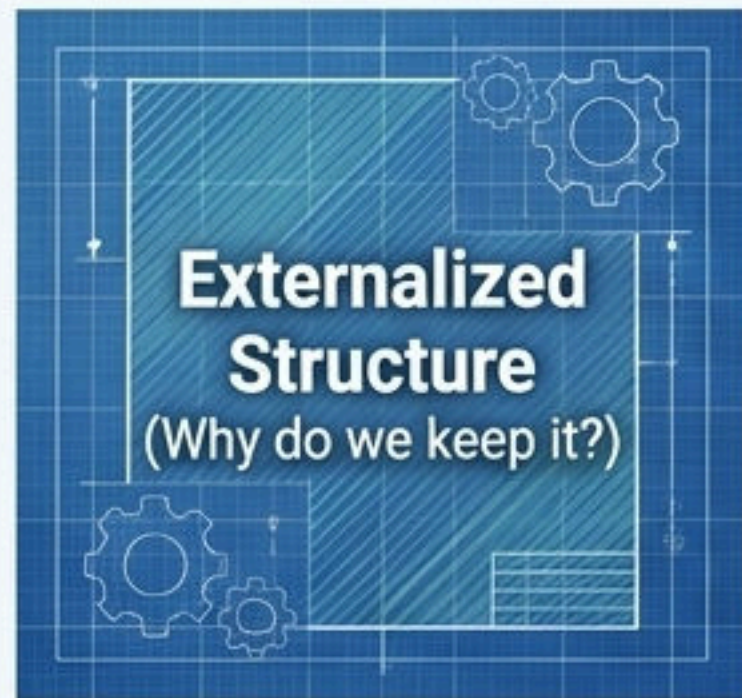
The False Inference

The public claimed finding a non-neural parser proved the AI possessed no real understanding—an illusion of a wizard behind a curtain.

The Rebuttal

Generative linguistics established long ago that syntactic computation possesses explanatory autonomy. Finding a parser is evidence of mature engineering, not a lack of semantic competence.

The Real Question: Why Does Structure Externalize?



The Standard Answer
(Incomplete)

Economics

A structure is kept because recomputing it is more expensive than storing it. (e.g., A cached parser tree).

The Neglected Reality

Topology

A structure is kept because, absent that structure, a prior state of the system's history becomes permanently unreachable from its current state. (e.g., A revision history log).

The Two Kinds of Persistence

Reconstruction-Cost Materialization

Driver:

Computational Economics.

Condition:

A Generating Input exists (a state independent of execution history from which the structure can be reproduced, e.g., a source file on disk).

Contingency:

Tracks the cost curve. Could disappear as compute gets cheaper.

Irreversibility Materialization

Driver:

System Topology.

Condition:

No Generating Input exists. The transition is Reconstructively Irreversible.

Contingency:

Absolute necessity. No improvement in hardware speed dissolves this requirement, because it is a claim about history, not cost.

Mechanics of Information Loss: Fiber Collapse



Transition System

A system defined by a transition function. Editing a file, revising a belief, or compacting context.

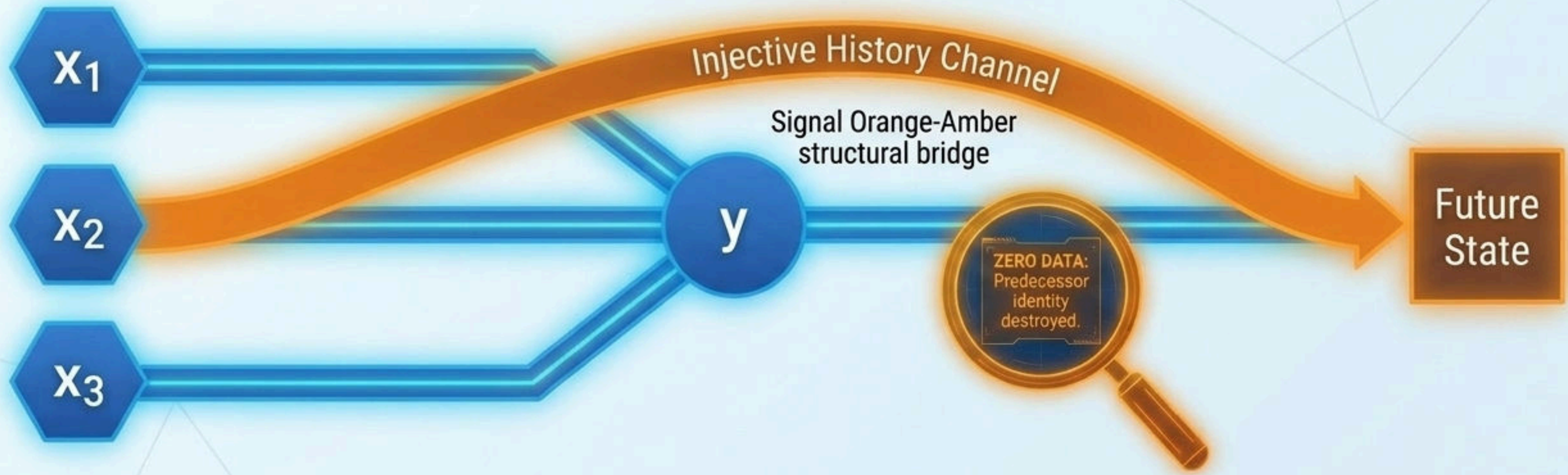
Fiber Collapse

Non-injectivity is the ordinary case—transitions generically collapse fibers. Indistinguishable inputs yield a single output.

Proposition 4.6

If a function is non-injective over a fiber, no function of the successor state y **alone determines which predecessor produced it.** The predecessor is not just expensive to reconstruct. It is destroyed.

Formalizing Irreversibility: History Channels



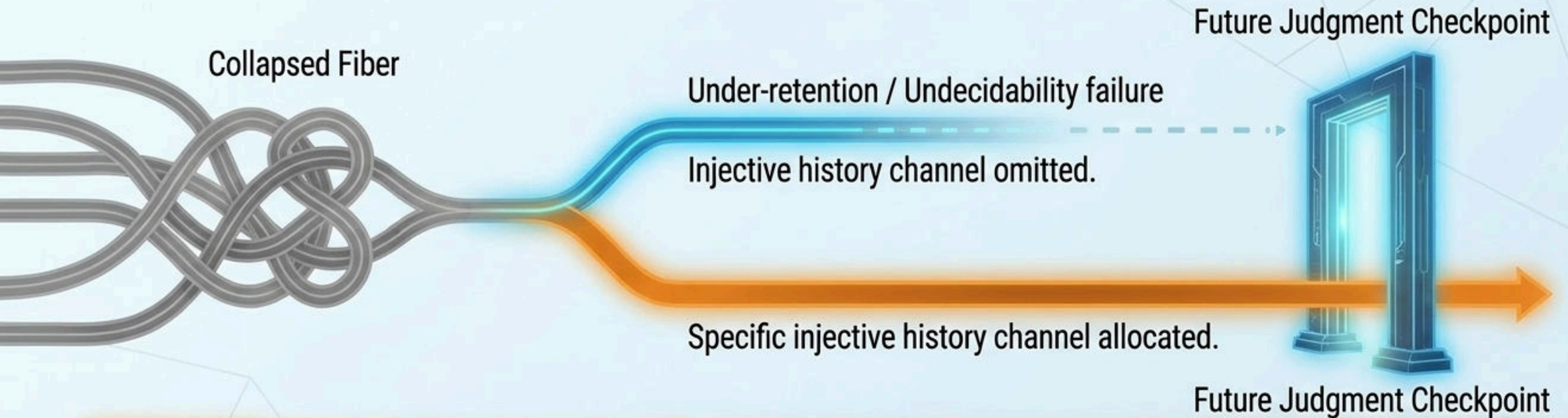
Historical Distinction Theorem (Theorem 4.9)

- To repair a collapsed fiber, a system augments the state with a history channel.
- The identity of the predecessor is recoverable from the current state if and only if the history channel is injective over the specific fiber that collapsed.

The Trap of Mere Retention

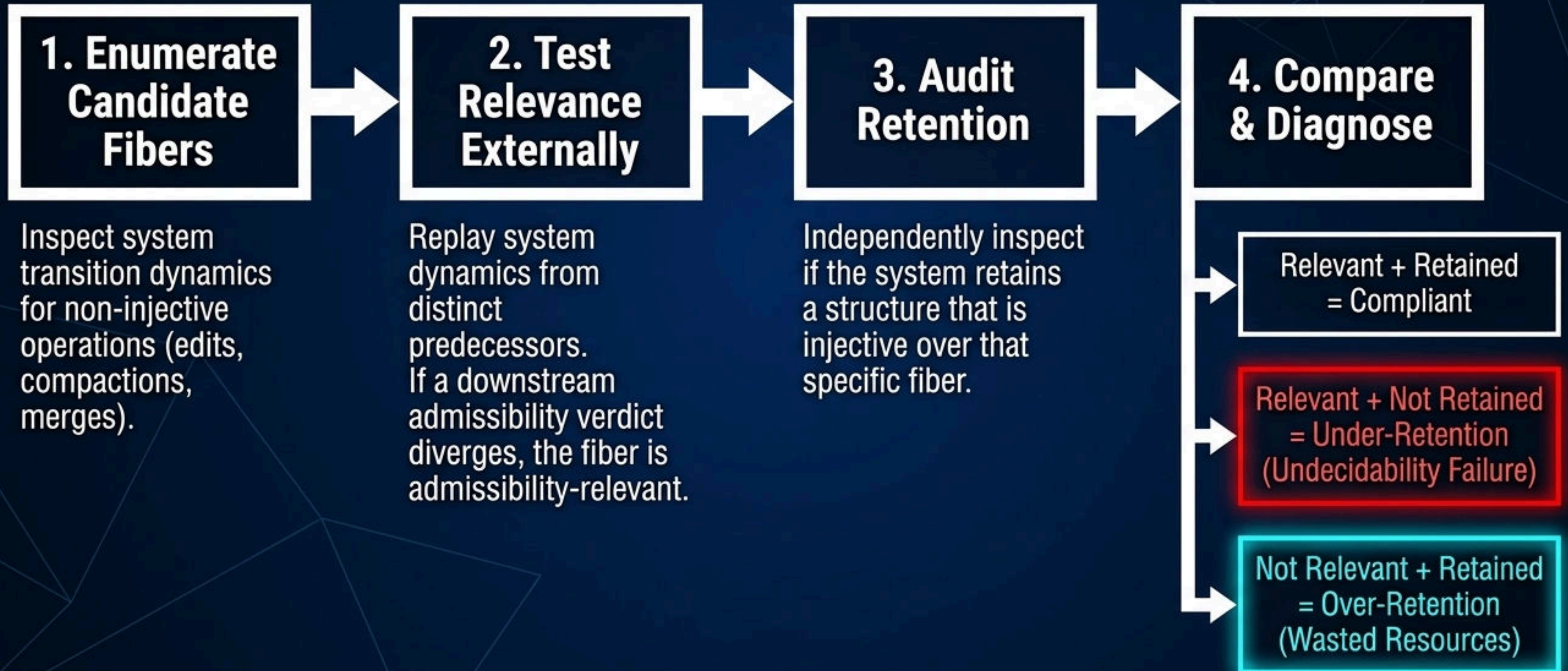
- A log recording "a file was edited" is non-injective (useless for recovery).
- A log recording the diff is injective.
- The mere presence of a trace does not entail it preserves distinguishability.

The Selective Retention Theorem: What Actually Matters



- A system cannot preserve distinguishability over every non-injective transition. It would drown in its own history.
- **Admissibility-Relevant Fibers:** A fiber matters only if some future admissibility judgment depends on which predecessor actually occurred.
- **Theorem 5.10 (Selective Retention):** A system minimizes retention cost by allocating injective history channels precisely over admissibility-relevant fibers, and discarding distinguishability everywhere else.

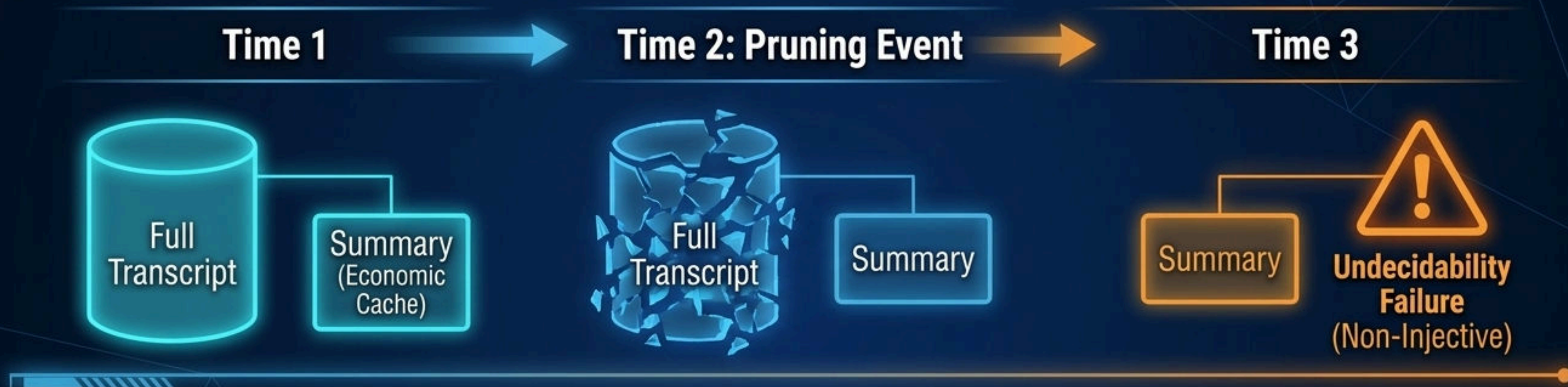
The Falsifiability Audit: How to Test the Theory



Case Study: Auditing the Claude Code Leak

Component	Generating Input?	Admissibility-Relevant?	Retention Observed?	Final Classification
Parse/Search Infrastructure	Yes (source disk)	No	Cached	Reconstruction-Cost
Pre-Risky-Operation Checkpoint	Yes (explicitly created)	Yes (rollback depends on it)	Retained	Irreversibility
Context Compaction Chain	Yes (while transcript lives)	Yes	Retained, boundaries patched	Ambiguous (Time-Dependent)
Cross-Session Trust State	No	Contested	Not Retained	Failure Mode Identified

The Decay of Generating Inputs: Checkpoints vs. Compaction



Confirmation (Checkpoints)

Checkpoints strictly confirm the theorem. Triggered specifically before risky operations, deliberately aligning retention with admissibility-relevance.

Ambiguity (Context Compaction)

A summarized context is **Reconstruction-Cost** only as long as the raw transcript survives.

The Silent Reclassification

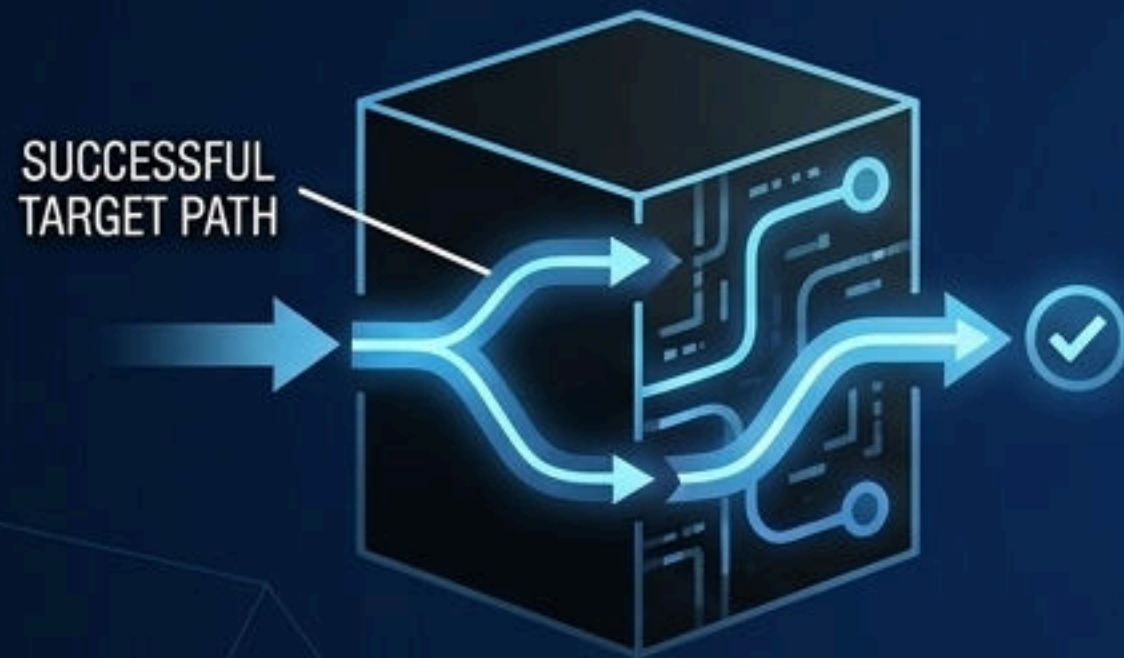
When memory limits prune the raw transcript, the summary silently becomes **Irreversibility Materialization**.

Because the summary is inherently lossy (non-injective), it fundamentally fails to preserve admissibility, causing silent undecidability failures downstream.

The Transformer Semantics Reframing

The “Logical Form” debate conflated two entirely different questions.

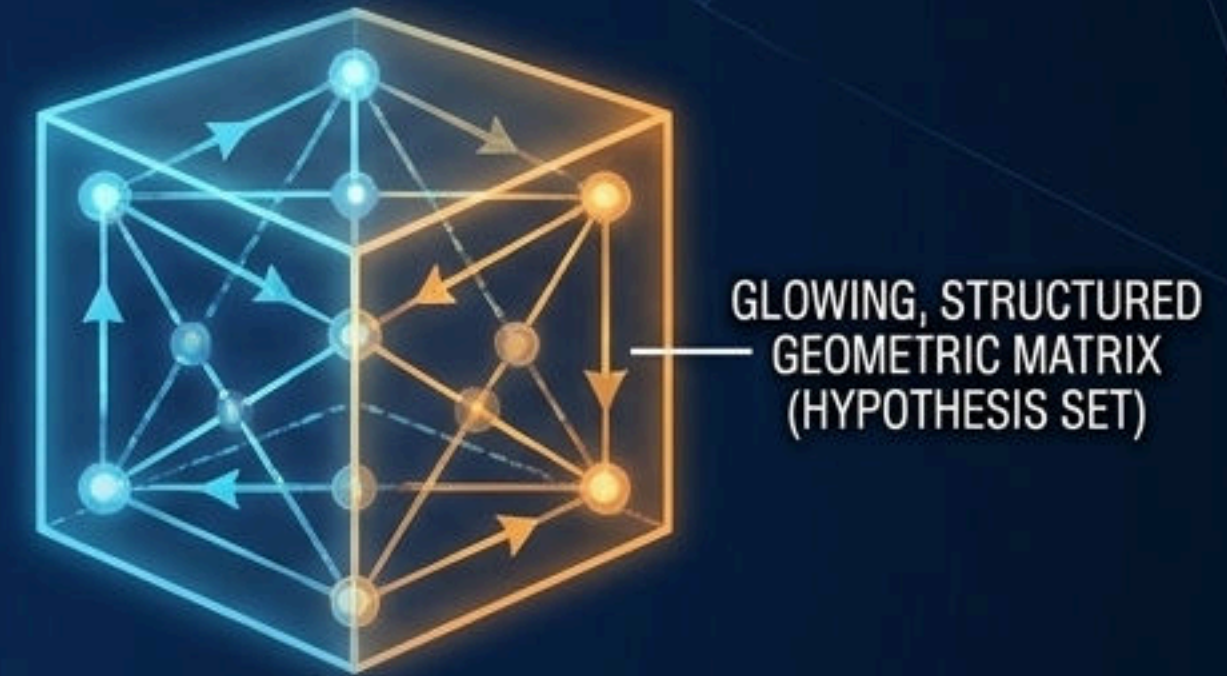
1. Functional Adequacy



The Test: Do downstream outputs track the true predecessor at better than chance across counterfactual contexts (e.g., distinguishing quantifier scopes)?

The Meaning: The system behaviorally resolves the collapsed fiber.

2. Architectural Transparency



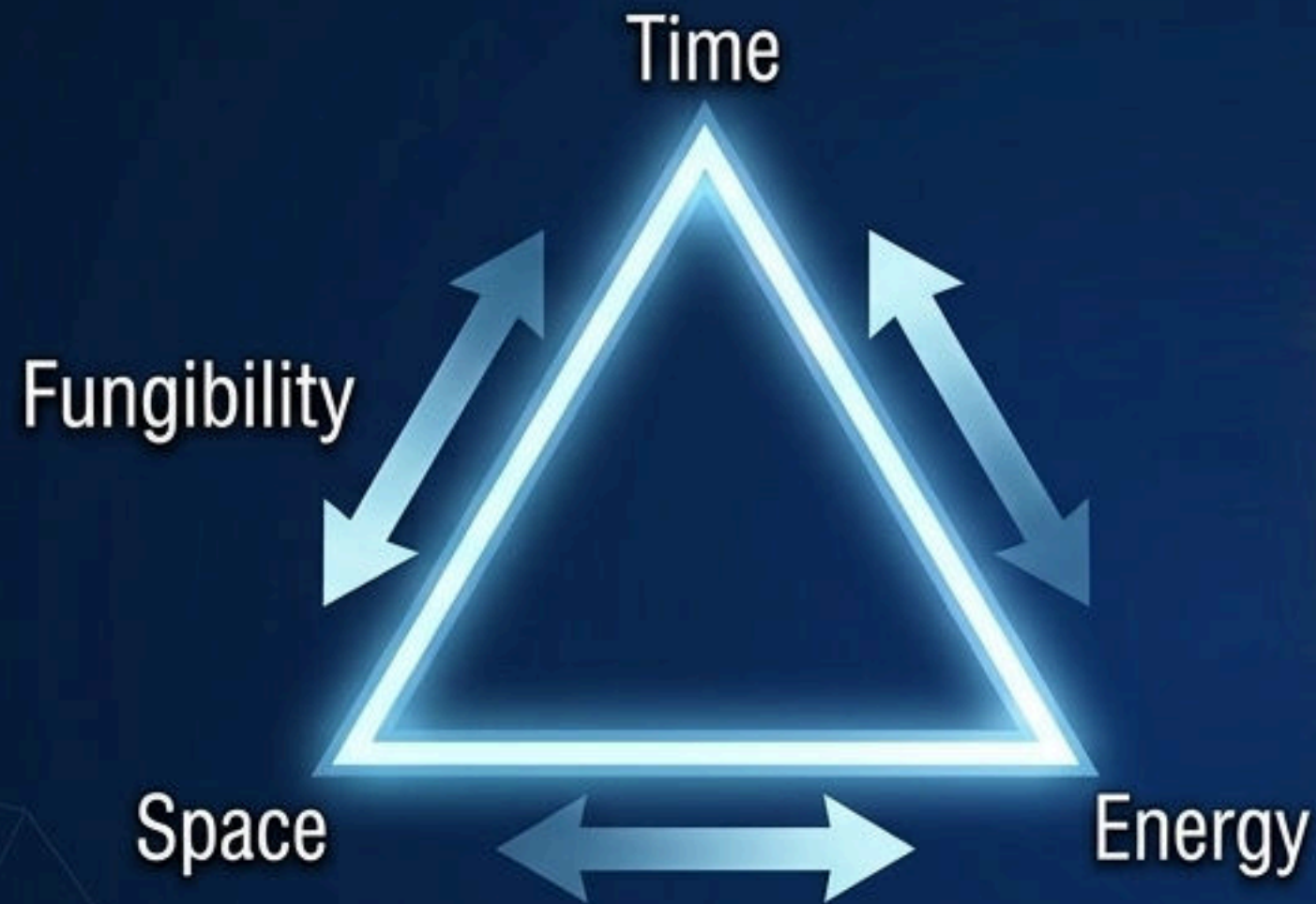
The Test: Does the system maintain an explicit, inspectable object (a hypothesis set) whose state can be compared against the fiber without probing latent vectors?

The Meaning: The system is engineered to track its own admissibility, matching HYDRA-level explicit representation. A model can be functionally adequate without being architecturally transparent.

The Physics of Persistence



Continuation as the Fourth Computational Resource



Time, Space, and Energy are fungible. You can trade memory for compute, or energy for speed.

The Resource Independence Theorem: If a generating input is destroyed, and the transition was restructively irreversible, no allocation of Time, Space, or Energy can recover the predecessor's identity.

Continuation is not "more memory." Unbounded memory cannot save a system that applied a non-injective retention policy.

Historical Observability is an independent dimension of computation.

The Admissible Externalization Conjecture

